



COMPONENTS

5G chip

MediaTek has announced Dimensity 820 system-on-chip (SoC) that delivers ultra-fast 5G speed and is feature-packed with the latest multimedia, artificial intelligence (AI), and imaging innovations, offering enhanced user experience. It is equipped with high-performance Arm Mali G57 GPU graphics and HyperEngine 2.0 enhancements that offer an extraordinary gaming experience on mobile phones.

The Dimensity 820 integrates a 5G NR (sub-6GHz) modem into this single-chip solution with MediaTek's 5G UltraSave technologies, including exclusive 5G UltraSave network environment detection and 5G UltraSave OTA content awareness that altogether provides best-in-class energy efficiency. The 5G modem design also uses dynamic bandwidth part (BWP) adaptation and connected mode discontinuous reception (C-DRX) power-saving technologies.

It supports dual SIM, dual standby (DSDS) 5G technology, and voice over new radio (VoNR) services on both SIMs. It has been designed for global sub-6GHz 5G networks in Asia, North America, and Europe.

MediaTek

www.mediatek.com

DC-DC converter

Xilinx has launched a 20-nanometer (nm) space-grade FPGA, the Kintex-UltraScale XQRKU060 that delivers full radiation tolerance, ultra-high throughput, and bandwidth performance for satellite and space applications. It also provides true unlimited on-orbit reconfiguration, which is ideal for payload applications and full radiation tolerance across all orbits.

The launch of XQRKU060 for space

applications delivers a significant reduction in size, weight, and power. It is engineered with robust radiation-tolerant features. The XQRKU060 provides customers with a space-resilient device equipped to handle both short and long duration missions in harsh space environments.

It is equipped with 2,760 UltraScale DSP processors and provides up to 1.6 TeraMACs of signal processing computing as well as dramatic efficiency gains for floating-point computations. The increased compute capability in space is paired with massive I/O bandwidth from 32 high-speed transceivers (SerDes) that can run up to 12.5Gbps to deliver 400Gbps aggregate bandwidth.

Xilinx

www.xilinx.com

IoT eSim

Cellular connectivity allows easy deployment and management of Internet of Things (IoT) devices by businesses due to its progressive infrastructure. To enable even better connectivity, OPTIGA



Connect eSIM has been launched by Infineon for cellular-connected IoT devices. Its small size aims to provide flexibility to device makers and manufacturers.

It enables hundred times faster data transmission and 1000 times greater data capacity coupled with heavy data encryption and mutual authentication of devices and networks at the same time. It is available in two forms - OPTIGA Connect Consumer and OPTIGA Connect IoT. Launched in March this year, the OPTIGA Connect Consumer allows consumer devices to connect to mobile network operators. It is 5G ready and provides security according to GSMA requirements.

The newest member in the OPTIGA Connect solution, OPTIGA Connect

IoT is based on leading-edge security hardware and comes pre-loaded with worldwide bootstrap connectivity.

Infineon

www.infineon.com

MCU with Bluetooth 5.0

Renesas Electronics has launched the first RA microcontroller (MCU) with an integrated Bluetooth 5.0 low energy radio. The single-chip RA4W1 MCU includes a 48 MHz, 32-bit Arm Cortex-M4 core, and Bluetooth 5.0 core delivered in a 56-pin QFN package. Together, the RA4W1 MCU and easy-to-use flexible software package (FSP) enable engineers to immediately begin development with Arm ecosystem software and hardware building blocks that work out-of-the-box with RA MCUs.



The RA4W1 MCU makes it easy for embedded designers to develop safe and secure IoT endpoint devices for Industry 4.0, building automation, metering, healthcare, consumer wearable, and home appliance applications. The MCU is also ideal for developing IoT edge devices for wireless sensor networks, IoT hubs, and an add-on to gateways.

The RA4W1 MCU features field upgradeable 512kB flash memory, 96 kB SRAM, and connectivity such as USB, CAN, and HMI capacitive touch technology.

It includes full Bluetooth 5.0 functions such as 2Mbps data throughput, all advertising extension functions with maximum advertising length (1650 byte), periodic advertisements, and channel selection algorithm #2 for applications requiring large traffic.

Renesas

www.renesas.com

Motion sensor

TDK Corporation has launched a high-performance motion sensor, the InvenSense ICM-42688-P. It is a 6-axis MEMS